

# 1 Chapter 1

All:1.1,1.2,1.7,1.14 and 6 examples

1.1:

$$x = \tan\theta d \quad \dot{x} = \frac{\dot{\theta}}{\cos^2\theta} d = \frac{\omega}{\cos^2\theta} d$$

$$\cos\theta = \frac{d}{\sqrt{d^2 + x^2}} \Rightarrow \frac{\omega(d^2 + x^2)}{d}$$

$$\ddot{x} = \frac{2x\dot{x}\omega}{d} = \frac{2x\omega}{d} \frac{\omega(d^2 + x^2)}{d} = \frac{2x\omega^2(d^2 + x^2)}{d^2}$$

1.2:

$$\begin{cases} x = \sin\theta b \\ y = \cos\theta d \end{cases}$$

$$\frac{d}{dt} \cos\theta(b + d) = -c$$

$$\Rightarrow \dot{\theta} = \frac{c}{\sin\theta(b + d)}$$

$$\begin{cases} \dot{x} = \frac{cb\cos\theta}{\sin\theta(b+d)} \\ \dot{y} = -\frac{dc}{(b+d)} \end{cases}$$

$$\begin{cases} \ddot{x} = -\frac{c^2 b}{\sin^3\theta(b+d)^2} \\ \ddot{y} = 0 \end{cases}$$

1.7:

$$\vec{v} = v_x \vec{i} + v_y \vec{j} \quad v_x^2 + v_y^2 = c$$

$$\vec{a} = \dot{v}_x \vec{i} + \dot{v}_y \vec{j}$$

$$\vec{a} \cdot \vec{v} = v_x \dot{v}_x + v_y \dot{v}_y = \frac{1}{2}(\dot{v}_x^2 + \dot{v}_y^2) = \frac{1}{2}\dot{c} = 0$$

1.14:

$$\begin{aligned}
& \begin{cases} m\ddot{x} = -mk\dot{x} \\ m\ddot{y} = -mk\dot{y} - mg \end{cases} \\
& \begin{cases} \dot{x} = C_1 e^{-kt} \\ \dot{y} = C_2 e^{-kt} - \frac{g}{k} \end{cases} \\
& t=0 \quad C_1 = v_{x0}; C_2 = \frac{g}{k} + v_{y0} \\
& -\tan\alpha = \frac{(\frac{g}{k} + v_{y0})e^{-kt} - \frac{g}{k}}{v_{x0}e^{-kt}} \\
& -\tan\alpha = (\frac{g}{kv_{x0}} + \tan\alpha) - \frac{g}{kv_{x0}}e^{kt} \\
& \Rightarrow e^{kt} = \frac{kv_{x0}}{g}(\frac{g}{kv_{x0}} + 2\tan\alpha) \\
& t = \frac{1}{k}\ln[\frac{kv_{x0}}{g}(\frac{g}{kv_{x0}} + 2\tan\alpha)] = \frac{1}{k}\ln[1 + 2\frac{k}{g}\sin\alpha v_0]
\end{aligned}$$

exp1:

$$\begin{aligned}
v &= \begin{cases} v_x = v_0 \\ v_y = gt \end{cases} \\
a_n &= \frac{v^2}{\rho} \Rightarrow \rho = \frac{v^2}{a_n} \\
a_n &= \vec{a} \cdot \vec{n} = \sqrt{a^2 - (a \cdot v \frac{1}{|v|})^2} \\
\vec{a} &= g\vec{j}; \vec{a} \cdot \vec{v} = g^2 t \\
a_n &= \sqrt{g^2 - \frac{g^4 t^2}{v_0^2 + g^2 t^2}} = \sqrt{\frac{v_0^2 g^2}{v_0^2 + g^2 t^2}} = \frac{v_0 g}{\sqrt{v_0^2 + g^2 t^2}} \\
\rho &= \frac{(v_0^2 + g^2 t^2)^{\frac{3}{2}}}{v_0 g}
\end{aligned}$$

exp2:

$$\begin{aligned}
m\ddot{y} &= -mg - mk\dot{y} \\
\ddot{y} + k\dot{y} &= -g \Rightarrow y = A + Be^{-kt} - \frac{gt}{k} \\
\dot{y}|_{t=0} &= 0 \Rightarrow \dot{y} = \frac{g}{k}e^{-kt} - \frac{g}{k}
\end{aligned}$$

exp3:

$$l = r \times p = 2 \times 0.5 \times 4 = 1 = v_B \times 0.5 \times 2 \\ \Rightarrow v_B = 1$$

exp4:

$$mg = kx \Rightarrow x = \frac{mg}{k} \\ \frac{1}{2}mv^2 - mgx + \frac{1}{2}kx^2 = 0 \\ \Rightarrow \frac{1}{2}mv^2 = \frac{m^2g^2}{k} - \frac{1}{2}k\frac{m^2g^2}{k^2} = \frac{m^2g^2}{2k} \\ v = \sqrt{\frac{m}{k}}g$$

exp5:

$$V = - \int (ax + by^2)dx + (az + 2bxy)dy + (ay + bz^2)dz \\ = - \frac{1}{2}ax^2 - \frac{1}{3}bz^3 - \int by^2dx + azdy + bxdy^2 + aydz \\ = - \frac{1}{2}ax^2 - \frac{1}{3}bz^3 - by^2x - azy + C \\ C = 0$$

exp6:

$$\frac{d^2x}{dt^2} = 2 + 6x^2 \\ \int_0^x adx = 2x + 2x^3 = \int_0^x \frac{d^2x}{dt^2}dx = \int_0^x vdv = \frac{1}{2}v^2 - \frac{1}{2}v_0^2 \\ v = \sqrt{100 + 4x + 4x^3}$$

## 2 Chapter 2

2.8,2.13,2 examples

2.8:

$$\vec{r} = r\vec{e}_r; \vec{v} = \dot{r}\vec{e}_r + r\omega\vec{e}_\theta;$$

$$\vec{a} = \ddot{r}\vec{e}_r + \dot{r}\omega\vec{e}_\theta + (\dot{r}\omega + r\dot{\omega})\vec{e}_\theta - r\omega^2\vec{e}_r = (\ddot{r} - r\omega^2)\vec{e}_r + (2\dot{r}\omega + r\dot{\omega})\vec{e}_\theta$$

$$r \times p = h \Rightarrow r^2\omega = \frac{h}{m}$$

$$\dot{r} = \frac{dr}{dt} = \frac{dr}{d\theta}\omega = \frac{dr}{d\theta}\frac{h}{mr^2} = -\frac{h}{m}\frac{d\frac{1}{r}}{d\theta}$$

$$r = ae^{b\theta} \Rightarrow \dot{r} = -\frac{h}{m}\frac{1}{a}(-b)e^{-b\theta} = \frac{bh}{m}\frac{1}{r}$$

$$\ddot{r} = -\frac{hb}{m}\frac{1}{r^2}\left(\frac{hb}{m}\frac{1}{r}\right) = -\frac{h^2b^2}{m^2}\frac{1}{r^3}$$

$$\ddot{r} - r\omega^2 = \frac{F}{m} = -\frac{k}{mr^3} = -\frac{h^2b^2}{m^2}\frac{1}{r^3} - r\frac{h^2}{m^2r^4}$$

$$\Rightarrow k = \frac{h^2b^2}{m} + \frac{h^2}{m} \Rightarrow h = \sqrt{\frac{km}{b^2 + 1}}$$

$$E = T + V = \frac{1}{2}m(\dot{r}^2 + r^2\omega^2) - \frac{k}{2r^2} = \frac{1}{2}m\left(\frac{h^2b^2}{m^2r^2} + \frac{h^2}{m^2r^2}\right) - \frac{k}{2r^2}$$

$$\Rightarrow \frac{1}{2r^2}\left(\frac{h^2b^2}{m} + \frac{h^2}{m} - k\right) = 0$$

2.13:

$$V = \frac{mc}{2r^2}$$

$$\frac{1}{2}mv_\infty^2 = \frac{1}{2}mv^2 + \frac{mc}{2d^2}$$

$$\rho v_\infty = vd \Rightarrow v = \frac{\rho v_\infty}{d}$$

$$= v_\infty^2 - \frac{\rho^2 v_\infty^2}{d^2} = \frac{c}{d^2}$$

$$d = \sqrt{\frac{c + \rho^2 v_\infty^2}{v_\infty^2}}$$

exp1:

$$\begin{aligned}
3av_0 &= av \Rightarrow v = 3v_0 \\
\frac{2k}{3a} &= 4mv_0^2 \Rightarrow k = 6amv_0^2 \\
\frac{1}{2}mv_0^2 + \frac{k}{6a} &= \frac{1}{2}mv^2 \\
\Rightarrow \frac{3}{2}mv_0^2 &= \frac{1}{2}mv^2 \\
v &= \sqrt{3}v_0
\end{aligned}$$

exp2:

$$\begin{aligned}
l &= \cos\alpha 2Rmv_0 = mvR \\
v &= 2\cos\alpha v_0 \\
\frac{mv_0^2}{2R} &= \frac{k}{4R^2} \Rightarrow k = 2Rmv_0^2 \\
\frac{3}{2}mv_0^2 &= \frac{1}{2}mv^2 \\
v &= \sqrt{3}v_0 \\
\cos\alpha &= \frac{\sqrt{3}}{2} \Rightarrow \alpha = \frac{\pi}{6}
\end{aligned}$$

### 3 Chapter 3

2 examples

exp1:

$$\begin{aligned}
r &= r_c + r'; v = \dot{r}' + \omega \times r'; a = \ddot{r}' + \omega \times \dot{r}' + \omega \times \omega \times r' + \omega \times \dot{r}' \\
r &= r' = b\vec{i} + (b - ut)\vec{j} \\
\vec{v} &= -u\vec{j} + \omega((b - ut)\vec{i} - b\vec{j}) \\
\vec{a} &= 0 - 2\omega u\vec{i} - 2\omega^2(b - ut)\vec{j} - 2b\omega^2\vec{i} + \omega \times \omega(-b\vec{j} + (b - ut)\vec{i}) \\
&= -2\omega u\vec{i} - 2\omega^2(b - ut)\vec{j} - 2b\omega^2\vec{i} - \omega^2 b\vec{i} - \omega^2(b - ut)\vec{j} \\
|v| &= \sqrt{\omega^2(b - ut)^2 + (\omega b + u)^2} \\
|a| &= \sqrt{(-\omega^2 b - 2b\omega^2 - 2\omega u)^2 + (-2\omega^2(b - ut) - \omega^2(b - ut))^2}
\end{aligned}$$

3.5:

$$m\vec{a}' = -mg\vec{j} - ma\vec{i} + \vec{T}$$

$$\begin{cases} ma_1 = \sin\alpha mg - ma_0 \cos\alpha \\ T = \cos\alpha mg + \sin\alpha ma_0 \end{cases}$$

## 4 Chapter 4

4.1 eg.4.2 and 4 examples

4.1:

$$x = r \sin\theta; y = r \cos\theta$$

$$\begin{cases} \bar{x} = \frac{\int x dm}{\int dm} = 0 \\ \bar{y} = \frac{\int y dm}{\int dm} \\ \bar{y} = \frac{2r^2 \sin \frac{s}{2r}}{s} \end{cases}$$

eg.4.2:

$$L = L_c + L'$$

$$\omega r = \Omega(R - r)$$

$$L_c = \Omega(R - r)^2 3m\vec{e}_1$$

$$L' = -3mr^2\omega\vec{e}_1$$

$$L = 3m\omega(rR - 2r^2)$$

$$T = T_c + T'$$

$$T_c = \frac{1}{2}3m\omega^2 r^2$$

$$T' = \frac{1}{2}3m\omega^2 r^2$$

$$T = 3m\omega^2 r^2$$

exp1:

$$T = T_c + T'$$

$$T_c = \frac{1}{2}4mv_c^2$$

$$T' = \frac{1}{2}4mr^2\omega s^2$$

$$\omega r = v_c$$

$$T = 4mv_c^2$$

exp2:

$$\begin{aligned}
 mv_0 &= (M + m)v \Rightarrow v = \frac{m}{M + m}v_0 \\
 \frac{1}{2}mv_0^2 &= \frac{1}{2}(M + m)v^2 + \mu mgs \\
 \Rightarrow \frac{1}{2}mv_0^2 - \frac{1}{2}\frac{m^2}{M + m}v_0^2 &= \mu mgs \\
 \Rightarrow \frac{M}{2\mu g(M + m)}v_0^2 &= s
 \end{aligned}$$

exp3:

$$\begin{aligned}
 mv_0 &= mv_1 + mv_2 \Rightarrow v_2 = v_0 - v_1 \\
 \frac{1}{2}mv_0^2 &= \frac{1}{2}mv_1^2 + \frac{1}{2}mv_2^2 \\
 v_0v_1 &= v_1^2 \Rightarrow v_1 = \begin{cases} v_0 \\ 0 \end{cases} \\
 v_2 &= \begin{cases} 0 \\ v_0 \end{cases} \quad v_2 \neq 0 \\
 v_1 &= 0
 \end{aligned}$$

exp4:

$$\begin{aligned}
 l &= r \times p = lmv_0 \\
 lmv_0 &= L' + L_c \\
 mv_0 &= 3mv \Rightarrow v = \frac{1}{3}v_0 \\
 L_c &= \frac{2}{3}l3m\frac{1}{3}v_0 = \frac{2}{3}mlv_0 \\
 L' &= \frac{1}{3}l2m\omega\frac{1}{3}l + \frac{2}{3}lm\omega\frac{2}{3}l = \frac{2}{3}ml^2\omega \\
 \Rightarrow lmv_0 &= \frac{2}{3}mlv_0 + \frac{2}{3}ml^2\omega \\
 \Rightarrow \omega &= \frac{v_0}{2l}
 \end{aligned}$$

## 5 Chapter 5

5.8,5.10,5.11,5.17,5.21,5.22, eg.5.4 and 8 examples.

5.8:

$$\begin{aligned}
L^a &= J_b^a \omega^b \\
J_{ij} &= \begin{bmatrix} \int y^2 + z^2 dm & -\int xy dm & -\int zx dm \\ -\int yx dm & \int x^2 + z^2 dm & -\int yz dm \\ -\int zx dm & -\int yz dm & \int x^2 + y^2 dm \end{bmatrix} \\
dm &= \frac{m}{l} dz; x = y = 0 \\
J_{ij} &= \begin{bmatrix} \frac{1}{3} ml^2 & 0 & 0 \\ 0 & \frac{1}{3} ml^2 & 0 \\ 0 & 0 & 0 \end{bmatrix} \\
\vec{\omega} &\neq [ \\
\vec{L} &= \frac{1}{3} ml^2 \sin \theta \omega \vec{i}
\end{aligned}$$

5.10:

$$\begin{aligned}
df &= \mu g dm \quad dm = \frac{m}{\pi R^2} ds = \frac{m}{\pi R^2} r dr d\theta \\
M &= \int r \times df = \int \frac{m}{\pi R^2} r^2 dr d\theta = 2\pi \frac{m}{\pi R^2} \frac{1}{3} R^3 = \frac{2}{3} m \mu g R \\
M \Delta t &= J \Delta \omega \\
\Rightarrow \frac{2}{3} \mu m g R \Delta t &= \frac{1}{2} m R^2 \omega_0 \\
\Delta t &= \frac{3}{4} \frac{\omega_0 R}{\mu g}
\end{aligned}$$



5.11:

$$df = kv^2 ds = k\omega^2 x^2 l dx$$

$$M = \int r \times df = \int k\omega^2 x^3 l dx = \frac{1}{4} k\omega^2 d^4 l$$

$$M = J\beta$$

$$J = \int x^2 dm = \int x^2 \frac{m}{d} dx = \frac{1}{3} md^2$$

$$\frac{d\omega}{dt} = -\frac{3}{4m} k\omega^2 l d^2$$

$$\Rightarrow -\frac{d\frac{1}{\omega}}{dt} = -\frac{3}{4m} kld^2$$

$$\frac{1}{\omega_0} - \frac{2}{\omega_0} = -\frac{3}{4m} kld^2 t$$

$$t = \frac{4m}{3kld^2\omega_0}$$

5.17:

(i) omitted

$$(ii) h = \frac{1}{2} gt^2$$

$$\frac{1}{3} ml^2 \omega^2 = mgl \Rightarrow \omega = \sqrt{\frac{3g}{l}}$$

$$\Delta\theta = \omega \Delta t = \sqrt{\frac{3g}{l}} \sqrt{\frac{2h}{g}} = \sqrt{\frac{6h}{l}}$$

$$round = \frac{1}{2\pi} rad = \frac{1}{\pi} \sqrt{\frac{3h}{2l}}$$

5.21:

$$\begin{aligned}
\mathcal{L} &= \frac{1}{2}m\dot{x}^2 + \frac{1}{2}\frac{2}{5}mR^2\dot{\theta}^2 + \int \mu mg dx - \int \mu mg R d\theta \\
\frac{d}{dt}\left(\frac{\partial \mathcal{L}}{\partial \dot{q}}\right) &= \frac{\partial \mathcal{L}}{\partial q} \\
\begin{cases} m\ddot{x} = \mu mg \\ m\frac{2}{5}R^2\ddot{\theta} = -\mu mg R \end{cases} \\
\Rightarrow \begin{cases} \dot{x} = \mu gt \\ \dot{\theta} = \omega_0 - \frac{5}{2}\mu\frac{g}{R}t \end{cases} \\
\dot{x}|_t = \omega|_t R \\
\mu gt = \omega_0 R - \frac{5}{2}\mu gt \\
t = \frac{2}{7}\frac{\omega_0 R}{\mu g} \\
x = \frac{1}{2}\mu gt^2 = \frac{1}{2}\frac{4}{49}\frac{\omega_0^2 R^2}{\mu g} = \frac{2}{49}\frac{\omega_0^2 R^2}{\mu g}
\end{aligned}$$

5.22:

$$\begin{aligned}
F \cos \theta R &= M \\
\cos \theta &= \frac{h - R}{R} \\
M \Delta t &= \frac{2}{5}mR^2\omega_0 = FR\frac{h - R}{R}\Delta t \\
F \Delta t &= mv_0 \Rightarrow \Delta t = \frac{mv_0}{F} \\
v_0 &= \omega_0 R \\
(h - R) &= \frac{2}{5}R \\
h &= \frac{7}{5}R
\end{aligned}$$

eg.5.4:

$$J_{ij} = \begin{bmatrix} \int y^2 + z^2 dm & -\int xy dm & -\int zx dm \\ -\int yx dm & \int x^2 + z^2 dm & -\int yz dm \\ -\int zx dm & -\int yz dm & \int x^2 + y^2 dm \end{bmatrix}$$

$$dm = \frac{m}{a^3} dx dy dz$$

$$J_{ij} = \begin{bmatrix} \frac{2}{3}ma^2 & -\frac{1}{4}ma^2 & -\frac{1}{4}ma^2 \\ -\frac{1}{4}ma^2 & \frac{2}{3}ma^2 & -\frac{1}{4}ma^2 \\ -\frac{1}{4}ma^2 & -\frac{1}{4}ma^2 & \frac{2}{3}ma^2 \end{bmatrix}$$

$$J = e_{\omega a} J_b^a e_{\omega}^b$$

$$e_{\omega} = \frac{1}{\sqrt{3}} \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$$

$$J_b^a e^b = \frac{1}{\sqrt{3}} \begin{bmatrix} \frac{1}{6}ma^2 \\ \frac{1}{6}ma^2 \\ \frac{1}{6}ma^2 \end{bmatrix}$$

$$e_a J_b^a e^b = \frac{1}{3} \left( \frac{1}{2} ma^2 \right) = \frac{1}{6} ma^2$$

$$T = \frac{1}{2} \omega_a J_b^a \omega^b = \frac{1}{2} \omega^2 J = \frac{1}{12} ma^2 \omega^2$$

exp1:

$$\mathcal{L} = \frac{1}{2} m \dot{l}^2 + \frac{1}{2} m R^2 \omega^2 - (c - m g \sin \alpha) - f l + \int f R d\theta$$

$$\begin{cases} m \ddot{l} = m g \sin \alpha - f \\ \frac{1}{2} m R^2 \frac{d\omega}{dt} = f R \end{cases}$$

$$\ddot{l} = \dot{\omega} R$$

$$m g \sin \alpha - f = 2f$$

$$\frac{1}{3} m g \sin \alpha = f$$

exp2:

$$J_{ij} = \begin{bmatrix} \int y^2 + z^2 dm & -\int xy dm & -\int zx dm \\ -\int yx dm & \int x^2 + z^2 dm & -\int yz dm \\ -\int zx dm & -\int yz dm & \int x^2 + y^2 dm \end{bmatrix}$$

$$y = r \sin \theta; x = r \sin \theta, z = 0; dm = \frac{m}{\pi R^2} r dr d\theta$$

$$J_{ij} = \begin{bmatrix} \frac{1}{4} m R^2 & 0 & 0 \\ 0 & \frac{1}{4} m R^2 & 0 \\ 0 & 0 & \frac{1}{2} m R^2 \end{bmatrix}$$

$$\vec{\omega} = \begin{bmatrix} \frac{\sqrt{3}}{2} \omega \\ 0 \\ \frac{1}{2} \omega \end{bmatrix}$$

$$T = \frac{1}{2} \omega J \omega = \frac{1}{2} \omega^2 \begin{bmatrix} \frac{\sqrt{3}}{2} & 0 & \frac{1}{2} \end{bmatrix} \cdot \begin{bmatrix} \frac{\sqrt{3}}{8} m R^2 \\ 0 \\ \frac{1}{4} m R^2 \end{bmatrix} = \frac{1}{2} \omega^2 \left( \frac{3}{16} m R^2 + \frac{1}{8} m R^2 \right)$$

$$= \frac{5}{32} \omega^2 m R^2$$

exp3:

$$\frac{1}{2} M \omega^2 l^2 + \frac{1}{2} \frac{1}{3} m l^2 \omega^2 - M g l - m g \frac{1}{2} l = 0$$

$$(2M + m) g l = (M l^2 + \frac{1}{3} m l^2) \omega^2$$

$$\omega = \sqrt{\frac{(2M + m) g}{(M + \frac{1}{3} m) l}}$$

exp4:

$$y = \frac{1}{2} l \sin \theta \Rightarrow \omega = \frac{2}{\cos \theta l} v$$

$$\dot{y}^2 + \frac{1}{3} l^2 \omega^2 = g(1 - \sin \theta) l \Rightarrow \dot{y}^2 + \frac{1}{3} l^2 \frac{4v^2}{\cos^2 \theta l^2} = g(1 - \sin \theta) l$$

exp5:

$$\mathcal{L} = \frac{1}{2}m_1\dot{l}^2 + \frac{1}{2}m_2\dot{l}^2 + \frac{1}{2}\frac{1}{2}m\dot{l}^2 - (C - m_1gl) - \int \mu m_2 g dl$$

$$\ddot{l} = \frac{(m_1 - \mu m_2)g}{(m_1 + m_2 + \frac{1}{2}m)}$$

$$T_1 - m_1g = -m_1\ddot{l} \Rightarrow T_1 = m_1(-\frac{(m_1 - \mu m_2)g}{(m_1 + m_2 + \frac{1}{2}m)} + 1)$$

$$T_1 = m_1(\frac{(m_2 + \mu m_2 + \frac{1}{2}m)g}{m_1 + m_2 + \frac{1}{2}m})$$

$$m_2\ddot{l} = T_2 - \mu m_2g \Rightarrow T_2 = m_2(\ddot{l} + \mu g)$$

$$T_2 = m_2(\frac{(1 + \mu)m_1 + \frac{1}{2}\mu m}{m_1 + m_2 + \frac{1}{2}m})g$$

exp6:

$$\cos\theta g = \frac{v^2}{2R}$$

$$\frac{1}{2}mv^2 + \frac{1}{2}\frac{2}{5}mR^2\omega^2 = mg2R(1 - \cos\theta)$$

$$\omega = \frac{v}{2R}$$

$$\frac{11}{20}\cos\theta = (1 - \cos\theta)$$

$$\cos\theta = \frac{20}{31}$$

$$\theta = \arccos(\frac{20}{31})$$

exp7:

$$l = \frac{1}{3}ml^2\omega_0$$

$$\frac{1}{2}\frac{1}{3}ml^2\omega_0^2 = mg\frac{1}{2}l \Rightarrow \omega = \sqrt{\frac{3g}{l}}$$

$$\frac{1}{3}ml^2\omega_0 = \frac{4}{3}ml^2\omega \Rightarrow \omega = \frac{1}{4}\omega_0$$

$$(1 - \cos\theta) = \frac{1}{12}$$

$$\cos\theta = \frac{11}{12}$$

$$\theta = \arccos\left(\frac{11}{12}\right)$$

exp8:

$$m\omega^2\frac{1}{2}l = -\frac{1}{2}mg + F_1$$

$$l^2\omega^2 = g\frac{3}{2}l \Rightarrow \omega = \sqrt{\frac{3g}{2l}}$$

$$F_1 = \frac{5}{4}mg$$

$$M\beta = mgl\frac{\sqrt{3}}{4} = \frac{1}{3}ml^2\beta \Rightarrow \beta = \frac{3\sqrt{3}}{4}\frac{g}{l}$$

$$a_1 = \beta\frac{1}{2}l = \frac{3\sqrt{3}}{8}g$$

$$\frac{3\sqrt{3}}{8}gm = \frac{\sqrt{3}}{2}gm - F_1 \Rightarrow F_1 = \frac{\sqrt{3}}{8}mg$$

$$|F| = \sqrt{\frac{3}{64} + \frac{25}{16}}mg = \frac{\sqrt{103}}{8}mg$$

## 6 Chapter 6

6.1 6.7 6.10 and 6 examples.

6.1:

$$q = \theta$$

$$\frac{h-y}{h} = \frac{\frac{1}{2}l_0}{l}$$

$$h = \sin\theta l$$

$$\Rightarrow \sin\theta l - y = \frac{1}{2}l_0 \sin\theta \Rightarrow y = \sin\theta l - \frac{1}{2}l_0 \sin\theta$$

$$2\cos\theta r = l \Rightarrow y = \sin(2\theta)r - \frac{1}{2}\sin\theta l_0$$

$$\delta y = 2\cos 2\theta r - \frac{1}{2}\cos\theta l_0; \delta w = mg\delta y = 0$$

$$4\cos 2\theta r - \cos\theta l_0 = 0$$

$$0 = 8\cos^2\theta r - 4r - \cos\theta l_0 \Rightarrow \cos\theta = \frac{l_0 \pm \sqrt{l_0^2 + 4(8r \times 4r)}}{16r}$$

$$\cos\theta > 0 \Rightarrow \cos\theta = \frac{l_0 + \sqrt{l_0^2 + 128r^2}}{16r}$$

$$l = 2\cos\theta r$$

$$\Rightarrow l = \frac{l_0 + \sqrt{l_0^2 + 128r^2}}{8}$$

$$\Rightarrow 8l - l_0 = \sqrt{l_0^2 + 128r^2} \Rightarrow 64l^2 - 16ll_0 = 128r^2$$

$$l_0 = \frac{64l^2 - 128r^2}{16l} = \frac{4l^2 - 8r^2}{l}$$

exp1:

$$\vec{r}_A = \sin\phi l \vec{i} - \cos\phi l \vec{j}; F_A = F \vec{i}$$

$$\vec{r}_C = \frac{1}{2}\sin\phi l \vec{i} - \frac{1}{2}\cos\phi l \vec{j}; F_C = -mg \vec{j}$$

$$\delta w = (\cos\phi \delta\phi l \vec{i} + \sin\phi \delta\phi l \vec{j}) F \vec{i} + (\frac{1}{2}\sin\phi \delta\phi l \vec{j})(-mg \vec{j})$$

$$0 = \cos\phi F - mg \frac{1}{2}\sin\phi$$

$$\phi = \arctan \frac{2F}{mg}$$

exp2:

$$\begin{aligned}
 q &= \alpha \\
 L &= -\cot\alpha a + 2\cos\alpha l \\
 y &= -L\vec{j} \\
 \delta y &= -\frac{1}{\sin^2\alpha}a + 2\sin\alpha l = 0 \\
 \alpha &= \arcsin\left(\frac{a}{2l}\right)^{\frac{1}{3}}
 \end{aligned}$$

exp3:

$$\begin{aligned}
 r_a &= \sin\theta l \vec{j}; F_A = -mg \vec{j} \\
 r_B &= \cos\theta l \vec{i}; F_B = -F \vec{i} \\
 \delta w &= 0 = -mg \vec{j} \cos\theta l \vec{j} \delta\theta + \sin\theta \delta\theta F \\
 F &= \cot\theta mg
 \end{aligned}$$

6.7:With Lagrangian

$$\begin{aligned}
 \mathcal{L} &= \frac{1}{2}m_2\dot{l}^2 + \frac{1}{4}m_1\dot{l}^2 - (C - m_2gl) \\
 \Rightarrow m_2\ddot{l} + \frac{1}{2}m_1\ddot{l} &= m_2g \\
 \ddot{l} &= \frac{m_2g}{m_2 + \frac{1}{2}m_1}
 \end{aligned}$$

6.7:With Hamiltonian

$$\begin{aligned}
 \mathcal{L} &= \frac{1}{2}m_2\dot{l}^2 + \frac{1}{4}m_1\dot{l}^2 + m_2gl \\
 p_l &= \frac{\partial L}{\partial \dot{q}} = m_2\dot{l} + \frac{1}{2}m_1\dot{l} \Rightarrow \dot{l} = \frac{1}{m_2 + \frac{1}{2}m_1}p_l \\
 H &= \frac{1}{m_2 + \frac{1}{2}m_1}p_l^2 - \frac{1}{2m_2 + m_1}p_l^2 - m_2gl \\
 H &= \frac{1}{2m_2 + m_1}p_l^2 - m_2gl \\
 \dot{l} &= \frac{\partial H}{\partial p} = \frac{1}{m_2 + \frac{1}{2}m_1}p \\
 \dot{p} &= -\frac{\partial H}{\partial q} = m_2g \\
 \ddot{l} &= \frac{1}{m_2 + \frac{1}{2}m_1}m_2g
 \end{aligned}$$



6.10:With Lagrangian

$$\begin{aligned}
q &= \theta \\
v &= (R-r)\dot{\theta}; \omega r = (R-r)\dot{\theta} \\
L &= \frac{1}{2}mv^2 + \frac{1}{2}\frac{1}{2}mr^2\omega^2 - (C - mg\cos\theta(R-r)) \\
&= \frac{1}{2}m(R-r)^2\dot{\theta}^2 + \frac{1}{4}m(R-r)^2\dot{\theta}^2 + mg\cos\theta(R-r) \\
&= \frac{3}{4}m(R-r)\dot{\theta}^2 + mg\cos\theta(R-r) \\
&\Rightarrow \frac{3}{2}(R-r)\ddot{\theta} = -g\sin\theta; \sin\theta \sim \theta \\
\frac{3}{2}(R-r)\ddot{\theta} + g\theta &= 0 \\
\theta &= A\cos(\sqrt{\frac{2g}{3(R-r)}}t + \phi) \\
T = \frac{2\pi}{\omega} &= 2\pi\sqrt{\frac{3(R-r)}{2g}}
\end{aligned}$$

6.10:With Hamiltonian

$$\begin{aligned}
p &= \frac{\partial L}{\partial \dot{q}} = \frac{3}{2}m(R-r)^2\dot{\theta} \Rightarrow \dot{\theta} = \frac{2}{3m(R-r)^2}p \\
H &= p\dot{\theta} - L = p\dot{\theta} - \frac{3}{4}(R-r)^2\dot{\theta}^2 - mg\cos\theta(R-r) \\
&= \frac{1}{3m(R-r)^2}p^2 - mg\cos\theta(R-r) \\
\dot{p} &= -\frac{\partial H}{\partial q} = -mg\sin\theta(R-r) \\
\ddot{\theta} &= -\frac{2}{3(R-r)}g\sin\theta
\end{aligned}$$

exp4:Lagrangian

$$\begin{aligned}
 q &= \theta \\
 L &= \frac{1}{2} \frac{1}{3} m l^2 \dot{\theta}^2 - (C - \cos\theta mg \frac{1}{2} l) \\
 \Rightarrow \frac{1}{3} m l^2 \ddot{\theta} &= -\sin\theta \frac{1}{2} m g \\
 \ddot{\theta} + \frac{3}{2} \frac{g}{l} \theta &= 0 \Rightarrow \theta = A \cos(\sqrt{\frac{2l}{3g}} t + \phi) \\
 T &= 2\pi \sqrt{\frac{3g}{2l}}
 \end{aligned}$$

exp4:Hamiltonian

$$\begin{aligned}
 p &= \frac{\partial L}{\partial \dot{\theta}} = \frac{1}{3} m l^2 \dot{\theta} \Rightarrow \dot{\theta} = \frac{3p}{m l^2} \\
 H &= p \dot{\theta} - \frac{1}{6} m l^2 \dot{\theta}^2 - \cos\theta mg \frac{1}{2} l \\
 &= \frac{3p^2}{2m l^2} - \frac{1}{2} \cos\theta m g l \\
 \dot{p} &= -\frac{\partial H}{\partial q} = -\frac{1}{2} \sin\theta m g l \\
 \ddot{\theta} &= \frac{3}{l} (-\frac{1}{2} \sin\theta g) = -\frac{3}{2} \sin\theta \frac{g}{l} \\
 &\text{omitted....}
 \end{aligned}$$

exp5:Lagrangian

$$\begin{aligned}
 q &= l; \omega r = \dot{l} \\
 L &= \frac{1}{2} m \dot{l}^2 + \frac{1}{2} \frac{1}{2} m r^2 \omega^2 + m g \sin\theta l \\
 &= \frac{3}{4} m \dot{l}^2 + m g l \sin\theta \\
 \Rightarrow \frac{3}{2} m \ddot{l} &= m g l \cos\theta \\
 \ddot{l} &= \frac{2}{3} g l \cos\theta
 \end{aligned}$$

exp5:Hamiltonian

$$\begin{aligned}
 p &= \frac{3}{2}m\dot{l} \Rightarrow \dot{l} = \frac{2}{3m}p \\
 H &= p\dot{l} - \frac{3}{4}m\dot{l}^2 - mgl\sin\theta \\
 &= \frac{2}{3m}p^2 - \frac{p^2}{3m} - mgl\sin\theta \\
 &= \frac{p^2}{3m} - mgl\sin\theta \\
 \dot{p} &= -\frac{\partial H}{\partial q} = mgl\cos\theta \\
 \ddot{l} &= \frac{2}{3}gl\cos\theta
 \end{aligned}$$

exp6:Lagrangian

$$\begin{aligned}
 q &= l; r\omega = \dot{l} \\
 L &= \frac{1}{2}m_1\dot{l}^2 + \frac{1}{2}\frac{1}{2}m_1r^2\omega^2 + \frac{1}{2}m_2\dot{l}^2 - (c - m_2gl) \\
 &= \frac{3}{4}m_1\dot{l}^2 + \frac{1}{2}m_2\dot{l}^2 + m_2gl \\
 \frac{3}{2}m_1\ddot{l} + m_2\ddot{l} &= m_2g \\
 \ddot{l} &= \frac{m_2}{\frac{3}{2}m_1 + m_2}g
 \end{aligned}$$

exp6:Hamiltonian

$$\begin{aligned}
 p &= (\frac{3}{2}m_1 + m_2)\dot{l} \Rightarrow \dot{l} = \frac{1}{\frac{3}{2}m_1 + m_2}p \\
 H &= p\dot{l} - L = \frac{1}{\frac{3}{2}m_1 + m_2}p^2 - (\frac{3}{4}m_1 + \frac{1}{2}m_2)\frac{1}{(\frac{3}{2}m_1 + m_2)^2}p^2 - m_2gl \\
 &= \frac{1}{3m_1 + 2m_2}p^2 - m_2gl \\
 \dot{p} &= -\frac{\partial H}{\partial q} = m_2g \\
 \ddot{l} &= \frac{1}{\frac{3}{2}m_1 + m_2}m_2g
 \end{aligned}$$